Philip Jonathan

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In brief

Philip Jonathan is a semi-retired statistical modeller at the School of Mathematical Sciences at Lancaster University, UK. His current research is reported at www.lancs.ac.uk/∼jonathan. He has more than 35 years experience in statistical modelling of physical systems, particularly the ocean environment, and a background in applied mathematics and physics. He is particularly interested in characterising extreme ocean environments, and the development of methodology for good design and re-analysis of offshore and coastal infrastructure. He has published on the development of methods for large-scale non-stationary marginal extreme value modelling with multidimensional covariates, and non-stationary extensions of multivariate, spatial and time-series extreme value methods, and applied these widely. He has developed and refined estimators summarising the tails of distributions, including return values, associated values and environmental design contours in the presence of uncertainty. His long-standing interests include Bayesian uncertainty analysis for complex systems, Bayesian inference for optimal design and probabilistic inversion in remote sensing, multivariate prediction and computational statistics generally. He has also published on ion physics, chemical kinetics, multivariate analysis, manufacturing process optimisation, chemometrics and quantitative structure–activity relationships, single pixel cameras, wave radar measurement, time-series and change-point analysis, volcanology and ancient writing systems. Illustrative references include:

E. J. Watts, R. Rees, **P. Jonathan** et al. Afar triple junction fed by single, asymmetric mantle upwelling. Nat. Geosci. https://doi.org/10.1038/s41561-025-01717-0, 2025.

K. Ewans and **P. Jonathan** Uncertainties in estimating the effect of climate change on 100-year return value for significant wave height. Ocean Eng. 272:113840. arXiv:2212.11049, 2023.

J. P. Grainger, A. Sykulski, K. Ewans, H. F. Hansen and **P. Jonathan**. A multivariate pseudo-likelihood approach to estimating ocean wave direction models. J. R. Stat. Soc. C 118: 1373-1384, 2023.

S.Tendijck, E. Eastoe, J. Tawn, D. Randell and **P. Jonathan**. Modelling the extremes of bivariate mixture distributions with application to oceanographic data. J. Amer. Statist. Assoc. 118:1373–1384, 2023.

C. Roberts, O. Shorttle, K. Mandel, M. Jones, R. Ijzermans, B. Hirst and **P. Jonathan**. Enhanced monitoring of atmospheric methane from space with hierarchical Bayesian inference. Environ. Res. Lett. 17:064037, 2022. arXiv:2111.12486.

R. Shooter, J. A. Tawn, E. Ross and **P. Jonathan**. Basin-wide conditional spatial extremes for severe ocean storms. Extremes 24:241–265, 2021.

E. Ross, M. Kereszturi, M. van Nee, D. Randell and **P. Jonathan**. On the spatial dependence of extreme ocean storm sea states. Ocean Eng. 145:1–14, 2017.

D. Randell, K. Turnbull, K. Ewans and **P. Jonathan**. Bayesian inference for non-stationary marginal extremes. Environmetrics 27:439–450, 2016.

M. J. Jones, M. Goldstein, **P. Jonathan**, and D. Randell. Bayes linear analysis for Bayesian optimal experimental design. Journal of Statistical Planning and Inference, 171:115–129, 2015.

P. Jonathan, K. C. Ewans, and D. Randell. Non-stationary conditional extremes of northern North Sea storm characteristics. Environmetrics, 125:172–188, 2014.

R. Lee, **P. Jonathan**, and P. R. Ziman. Pictish symbols revealed as a written language through application of Shannon entropy. Proc. R. Soc. A, 467:309–313, 2011.

P. Jonathan, W. J. Krzanowski, and W. V. McCarthy. On the use of cross-validation to assess performance in multivariate prediction. Statistics and Computing, 10:209–229, 2000.

P. H. Taylor, **P. Jonathan**, and L. A. Harland. Time-domain simulation of jack-up dynamics using the extremes of a random process. Journal of Vibrations and Accoustics, 119:624, 1995.

M. Stone and **P. Jonathan**. Statistical thinking and technique for QSAR and related studies. part I : General theory. J. Chemometr. 7:455, 1993.

E. Y. Kamber, **P. Jonathan**, A. G. Brenton, and J. H. Beynon. Single-electron capture by Ar^{++} from atomic and molecular targets. J. Phys. B: Atom. Mol. Physics, 20:4129, 1987.

Employment

2018–date **Chair, Environmental Statistics and Data Science**, *Lancaster University*, Lancaster, UK.

2018–2025 **Chief Statistician**, *Shell Projects & Technology*, London, UK.

2004–2018 **Head, Statistics & Data Science**, *Shell Projects & Technology*, London, UK.

1996–2004 **Statistician**, *Shell Thornton*, Chester, UK.

1993–1996 **Mathematician**, *Shell Research*, Rijswijk, NL.

1992–1993 **Consulting Statistician**, *Shell Development Company*, Houston, USA.

1988–1992 **Mathematician**, *Shell Research*, Sittingbourne, UK.

1987–1988 **Process investigator**, *Morganite Electrical Carbon*, Swansea, UK.

Languages

Welsh Mother tongue.

English Fluent.

Czech Conversational.

Academic qualifications, honours, accreditations and affiliations

2021 **Greenfield Industrial Medallist**, *Royal Statistical Society, UK.*

2008–2018 **Honorary Research Fellow**, *Mathematics & Statistics*, Lancaster University, UK.

2000–2003 **Visiting Fellow**, *Chemical Engineering*, Newcastle University, UK.

1999 **Chartered Statistician**, *Royal Statistical Society*, UK.

1984–1987 **Ph.D.**, *Ion Physics*, Swansea University, UK.

1981–1984 **B.Sc.**, *Applied Mathematics*, Swansea University, UK, (Charles Prize in Mathematics, 1983; Olroyd Prize in Applied Mathematics, 1984).

Research interests

Extremes Extreme value analysis for meteorological-oceanographic applications.

Uncertainty Uncertainty quantification in models for complex systems.

Inversion Probabilistic inversion in remote sensing.

Academic supervision

2026–date **J. Guntrip**, *Ph.D., School of Mathematical Sciences, Lancaster University, UK.*

2022–2026 **M. Speers**, *Ph.D., Department of Mathematics & Statistics, Lancaster University, UK.*

2022–2025 **T. Newman**, *Ph.D., Department of Mathematics & Statistics, Lancaster University, UK.*

2021–2022 **D. Dodd**, *Department of Mathematics & Statistics, Lancaster University, UK.*

2019–2022 **J. Grainger**, *Ph.D., Department of Mathematics & Statistics, Lancaster University, UK.*

2019–2022 **S. Tendijck**, *Ph.D., Department of Mathematics & Statistics, Lancaster University, UK.*

2016–2020 **R. Shooter**, *Ph.D., Department of Mathematics & Statistics, Lancaster University, UK.*

2014–2018 **E. Zanini**, *Ph.D., Department of Mathematics & Statistics, Lancaster University, UK.*

2013–2017 **M. Kereszturi**, *Ph.D., Department of Mathematics & Statistics, Lancaster University, UK.*

2012–2017 **M. J. Jones**, *Ph.D., School of Mathematical Sciences, Durham University, UK.*

2012–2015 **B. Pickering**, *Ph.D., Department of Mathematics & Statistics, Lancaster University, UK.*

2011–2012 **D. Reeve**, *M.Res., Department of Mathematics & Statistics, Lancaster University, UK.*

2010–2015 **R. Towe**, *Ph.D., Department of Mathematics & Statistics, Lancaster University, UK.*

2010–2023 **Y. Wu**, *Ph.D., Department of Mathematics & Statistics, Lancaster University, UK.*

2009–2012 **J. Wadsworth**, *Ph.D., Department of Mathematics & Statistics, Lancaster University, UK.*

2008–2011 **R. Killick**, *Ph.D., Department of Mathematics & Statistics, Lancaster University, UK.*

2006–2010 **D. Randell**, *Ph.D., School of Mathematical Sciences, Durham University, UK.*

2003–2006 **G. Hardman**, *Ph.D., School of Mathematical Sciences, Durham University, UK.*

2000–2003 **J. Little**, *Ph.D., School of Mathematical Sciences, Durham University, UK.*

1999–2002 **J. Badcock**, *Ph.D., School of Mathematical Sciences, Exeter University, UK.*

1997–2000 **E. Kaskavelis**, *Ph.D., Centre for Process Analytics and Control Technology, U. Newcastle, UK.*

Other academic activities

Talks and seminars: Oceanographic extremes (Norwegian Metoffice and DNV, 2024; RSS conference, 2023; UK Metoffice, 2022; Royal Statistical Society Conference, 2021; ISSC-ITTC, 2021; Waves SIG, 2020; OMAE Conference, 2020; Lancaster extremes workshop, 2019; Marine Centre Wales, Bangor, 2018; Extreme Value Analysis & Natural Hazards, Southampton, 2017); Remote sensing (Wales Mathematics Colloquium, Gregynnog, 2022; Data-centric engineering, 2021); Emerging themes in data science (Emerging Applications Section, RSS, London, 2017); Spatial extremes (Extreme Value Analysis & Natural Hazards, Southampton, 2017); Real-time data and applications (StatScale workshop, Lancaster, 2017).

Academic leadership: Member of REF 2021 Mathematics Sub-panel (2018-21); Advisory Board for Mathematical Sciences (Durham, 2016-2017); StatScale External Advisory Board (Mathematics and Statistics, Lancaster, 2016-2018); STOR-i External Advisory Board (Mathematics and Statistics, Lancaster, 2009-2018).

External examination of PhDs: Mathematical Sciences (Oslo, 2024); Mathematical Sciences (Durham, 2020); Civil Engineering and Environmental Science (Imperial College, 2017); Chemical Engineering and Advanced Materials, Newcastle, 2016); Mathematical Sciences (Trondheim, 2015); Transport Studies (Leeds, 2011); Process Analytics and Control Technology (Newcastle, 2003). Also acted as Internal Examiner for Lancaster PhDs on numerous occasions since 2018.

Consultancy activities

Since 2024 I have undertaken a limited amount of consultancy work.

JBA / Environment Agency: On behalf of Lancaster University, support for review and update of flood and coastal defence Joint Probability Guidance (2024-).

IOGP: On behalf of Lancaster University, masterclass on applied extreme value analysis for metocean design (2026).

Yn y Gymraeg

Modelydd ystadegol yn Ysgol Gwyddorau Mathemategol Prifysgol Caerhirfryn yw Philip Jonathan. Mae'n astudio sistemau amgylcheddol yn cynnwys eithafon stormydd morol a gwasgariad nwyon atmosfferig gan ddefnyddio dulliau gwerthoedd eithafol a gofodol-amserol Bayesaidd. Gweler `www.lancs.ac.uk/~jonathan`.

Publications

Publications are sorted first by year (in reverse), then by author and finally if necessary by title.

2026: P. Bohlinger, **P. Jonathan**, B. Youngman, T. Economou, J.-V. Björkqvist, Ø. Breivik. Modelling marginal and joint extremes – introducing the EnvEx approach showcased for metocean related challenges (submitted to Ocean Eng. in September 2026, preprint at `www.lancs.ac.uk/~jonathan`).

2026: M. Chung, R. Wada, J. Rohmer and **P. Jonathan**. Temporally coherent modeling of tropical cyclone compound flooding for reliable coastal hazard estimation (submitted to Nat. Hazards Earth Syst. Sci. in March 2026, preprint at `www.lancs.ac.uk/~jonathan`).

2026: C. Leach, K. Ewans and **P. Jonathan**. Estimating changes in extreme quantiles over time, applied to desert temperatures. Stoch. Environ. Res. Risk Assess. 40:216. arXiv:2603.07227.

2026: E. Mackay, J. Richards and **P. Jonathan**. Diagnostic tools for extreme value regression models (submitted to Extremes in June 2026). arXiv:2606.02676.

2026: M. Speers, J. A. Tawn and **P. Jonathan**. Sequential design for the efficient estimation of offshore structure failure probability. Ocean Eng. 349:123990. arXiv:2509.18319.

2026: E. Vanem, S. Dong, G. de Hauteclocque, T. B. Johannesen, T. Zhu, J. Prpic-Orsic, S. van Essen, K. Ewans, E. Mackay and **P. Jonathan**. Statistical modeling of the ocean environment. Annu. Rev. Stat. Appl. 13:1-28.

2025: C. Leach, K. Ewans and **P. Jonathan**. Changes over time in the 100-year return value of climate model variables. Ocean Eng. 324:120605. arXiv:2501.11650.

2025: E. Mackay, C. J. R. Murphy-Barltrop, J. Richards and **P. Jonathan**. Deep learning joint extremes of metocean variables using the SPAR model. J. Offshore Mech. Arct. Eng. 148:021201. arXiv:2412.15808.

2025: C. Murphy, R. Towe and **P. Jonathan**. Calibration of medium-range metocean forecasts for the North Sea. Appl. Ocean Res. 158:104538. arXiv:2503.21488.

2025: T. Newman, C. Nemeth, M. J. Jones and **P. Jonathan**. Probabilistic inversion modeling of gas emissions: a gradient-based MCMC estimation of Gaussian plume parameters. Ann. Appl. Stat. 19:2937-2956. arXiv:2408.01298.

2025: T. Newman, C. Nemeth, M. J. Jones and **P. Jonathan**. Deep learning surrogates for real-time gas emission inversion. arXiv:2506.14597.

2025: Rajpaul, V. M., Marr, C., **P. Jonathan** and Hughes, B. M. Economi Cymru: cyfraniad y gwyddorau mathemategol. Academi'r Gwyddorau Mathemategol. DOI: 10.5281/zenodo.17777109.

2025: R. Paleja, O. Ekhorutomwen, M. Jones, John Ayoola, R. Pitchumani and **P. Jonathan**. Reducing the CO_2 footprint at an LNG asset with replicate trains using operational data-driven analysis. A case study on end flash vessels. Data-Centric Engineering 6:e1 https://doi.org/10.1017/dce.2024.23.

2025: E. J. Watts, R. Rees, **P. Jonathan**, D. Keir, R. N. Taylor, M. Siegburg, E. L. Chambers, C. Pagli, M. J. Cooper, A. Michalik, J. A. Milton, T. K. Hincks, E. F. Gebru, A. Ayele, B. Abebe and T. M. Gernon. Afar triple junction fed by single, asymmetric mantle upwelling. Nat. Geosci. https://doi.org/10.1038/s41561-025-01717-0.

2024: E. B. L. Mackay, C. J. R. Murphy-Barltrop and **P. Jonathan**. The SPAR model: a new paradigm for multivariate extremes. Application to joint distributions of metocean variables. J. Offshore Mech. Arct. Eng. 147:011205.

2024: C. J. R. Murphy-Barltrop, E. Mackay and **P. Jonathan**. Inference for multivariate extremes via a semi-parametric angular-radial model. Extremes https://doi.org/10.1007/s10687-024-00492-2. arXiv:2401.07259.

2024: K. Sando, R. Wada, J. Rohmer and **P. Jonathan**. Multivariate spatial and spatio-temporal models for extreme tropical cyclone seas. Ocean Eng. 309:118365.

2024: M. Speers, D. Randell, J. A. Tawn and **P. Jonathan**. Estimating metocean environments associated with

extreme structural response to demonstrate the dangers of environmental contour methods. Ocean Eng. 311:118754. arXiv:2404.16775.

2024: S. Tendijck, **P. Jonathan**, D. Randell and J. A. Tawn. Temporal evolution of the extreme excursions of multivariate kth order Markov processes with application to oceanographic data. Environmetrics. 35:e2834. arXiv:2302.14501.

2024: S. Tendijck, D. Randell, G. Feld and **P. Jonathan**. Practical non-stationary extreme value analysis of peaks over threshold using the generalised Pareto distribution: estimating uncertainties in return values. Ocean Eng. 312:119247.

2024: R. Towe, E. Ross, D. Randell and **P. Jonathan**. covXtreme : MATLAB software for non-stationary penalised piecewise constant marginal and conditional extreme value models. Environ. Model. Softw. 177:106035. arXiv:2309:17295.

2023: A.M. Barlow, E. Mackay, E. Eastoe and **P. Jonathan**. A penalised piecewise-linear model for non-stationary extreme value analysis of peaks over threshold. Ocean Eng. 267:113265. arXiv:2201.03915.

2023: K. Ewans and **P. Jonathan** Is the 100-year return value for significant wave height increasing in the Tasman Sea? (Proc. 42th Int. Conf. on Ocean, Offshore & Arctic Engineering, Mebourne; preprint at www.lancs.ac.uk/∼jonathan.)

2023: K. Ewans and **P. Jonathan** Uncertainties in estimating the effect of climate change on 100-year return value for significant wave height. Ocean Eng. 272:113840. arXiv:2212.11049.

2023: J. P. Grainger, A. Sykulski, K. Ewans, H. F. Hansen and **P. Jonathan**. A multivariate pseudo-likelihood approach to estimating ocean wave direction models. J. R. Stat. Soc. C. 72:544-565. arXiv:2202.03773.

2023: E. Mackay and **P. Jonathan**. Modelling multivariate extremes through angular-radial decomposition of the density function. (Submitted to arXiv in December 2023, submitted to Extremes in February 2024) arXiv:2310.12711.

2023: C. Roberts, R. Ijzermans, D. Randell, M.Jones, **P. Jonathan**, K. Mandel, B. Hirst and O. Shorttle. Avoiding methane emission rate underestimates when using the divergence method. Environ. Res. Lett. 18:114033. arXiv:2304.10303.

2023: S.Tendijck, E. Eastoe, J. Tawn, D. Randell and **P. Jonathan**. Modelling the extremes of bivariate mixture distributions with application to oceanographic data. J. Amer. Statist. Assoc. 118:1373–1384.

2023: S. Tendijck, J. A. Tawn and **P. Jonathan**. Extremal characteristics of conditional models. Extremes 26:139–156. arXiv:2202.11673.

2023: R. Towe, D. Randell, J. Kensler, G. Feld and **P. Jonathan**. Estimation of associated values from conditional extreme value models. Ocean Eng. 272:113808.

2022: A. Raby, G. Bullock, **P. Jonathan**, D. Randell and C. Whittaker. On wave impact pressure variability. Coastal Eng. 177:104168.

2022: C. Roberts, O. Shorttle, K. Mandel, M. Jones, R. Ijzermans, B. Hirst and **P. Jonathan**. Enhanced monitoring of atmospheric methane from space with hierarchical Bayesian inference. Environ. Res. Lett. 17:064037. arXiv:2111.12486.

2022: K. Sando, R. Wada, J. Rohmer, S. Lecacheux and **P. Jonathan**. Estimating joint extremes of significant wave height and wind speed for tropical cyclones. OMAE2022-79888 Proc. 41st Int. Conf. on Ocean, Offshore & Arctic Engineering. (Preprint at www.lancs.ac.uk/∼jonathan.)

2022: R. Shooter, E. Ross, A. Ribal, I.R. Young and **P. Jonathan**. Multivariate spatial conditional extremes for extreme ocean environments. Ocean Eng. 247:110647. arXiv:2201.10451.

2022: R. Wada, J. Rohmer, Y. Krien and **P. Jonathan**. Statistical estimation of spatial wave extremes for tropical cyclones from small data samples: validation of the STM-E approach using long-term synthetic cyclone data for the Caribbean Sea. Nat. Hazards Earth Sys., 22, 431-444.

2021: P. A. Carling, **P. Jonathan** and S. Teng. Fitting Limit Lines (envelope curves) to spreads of geoenvironmental data. Prog. Phys. Geog.: Earth & Env. 46:272-290.

2021: J. P. Grainger, A. Sykulski, **P. Jonathan** and K. Ewans. Estimating the parameters of ocean wave spectra. Ocean Eng. 229:108934. arXiv:2008.10437.

2021: A. F. Haselsteiner, R. G. Coe, L. Manuel, W. Chai, B. Leira, G. Clarindo, C. Guedes Soares, A. Hannesdottir, N. Dimitrov, A. Sander, J.-H. Ohlendorf, K.-D. Thoben, G. de Hauteclocque, E. Mackay, **P. Jonathan**, C. Qiao, A. Myers,

A. Rode, A. Hildebrandt, B. Schmidt, E. Vanem, A. B. Huseby. A benchmarking exercise for environmental contours. Ocean Eng. 236:109504.

2021: **P. Jonathan**. Cynrychioliad amharamedrig ar gyfer cydnewidynnau aml-ddimensiynol mewn modelau gwerthoedd eithafol (A non-parametric representation for multi-dimensional covariates in an extreme value model, with oceanographic application), Gwerddon 33:68-84.

2021: **P. Jonathan**, D. Randell, J. Wadsworth and J. Tawn. On return values for extreme seas. Ocean Eng. 220:107725.

2021: M. J. Jones, M. Goldstein, D. Randell and **P. Jonathan**. Bayes linear analysis for ordinary differential equations. Comput. Statist. Data Anal. 161:107228.

2021: E. Konzen, C. Neves and **P. Jonathan**. Modelling non-stationary extremes of storm severity: a tale of two approaches. Environmetrics 32:e2667. arXiv:2005.13490.

2021: E. Mackay, **P. Jonathan**. Sampling properties and empirical estimates of extreme events. Ocean Eng. 239:109791.

2021: E. Mackay, G. de Hauteclocque, E. Vanem and **P. Jonathan**. The effect of serial correlation in environmental conditions on estimates of extreme events. Ocean Eng. 242:11092.

2021: R. Shooter, E. Ross, A. Ribal, I.R. Young and **P. Jonathan**. Spatial dependence of extreme seas in the North East Atlantic from satellite altimeter measurements. Environmetrics 32:e2674.

2021: R. Shooter, J. A. Tawn, E. Ross and **P. Jonathan**. Basin-wide conditional spatial extremes for severe ocean storms. Extremes 24:241–265.

2021: R. Towe, E. Zanini, D. Randell, G. Feld and **P. Jonathan**. Efficient estimation of distributional properties of extreme seas from a hierarchical description applied to calculation of un-manning and other weather-related operational windows. Ocean Eng. 238:109642.

2020: K. Ewans, M. Christou, S. Ilic and **P. Jonathan**. Identifying higher-order interactions in wave time-series. J. Offshore Mech. Arctic Eng. 143:021201-1.

2020: K. Ewans and **P. Jonathan**. Extreme conditions. In I.R. Young and A. Babanin, editors, Ocean wave dynamics. World Scientific, pp 271-319.

2020: H. F. Hansen, D. Randell, A. R. Zeeberg and **P. Jonathan**. Directional-seasonal extreme value analysis of North Sea storm conditions. Ocean Eng. 195:106665.

2020: E. Mackay and **P. Jonathan**. Assessment of return value estimates from stationary and non-stationary extreme value models. Ocean Eng. 207:107406.

2020: E. Mackay and **P. Jonathan**. Estimation of environmental coutours using a block resampling method. Proc. 39th Int. Conf. on Ocean, Offshore & Arctic Engineering.

2020: E. Ross, O. C. Astrup, E. Bitner-Gregersen, N. Bunn, G. Feld, B. Gouldby, A. Huseby, Y. Liu, D. Randell, E. Vanem and **P. Jonathan**. On environmental contours for marine and coastal design. Ocean Eng. 195:106194. arXiv:1812.07886.

2020: M. Schubert, Y. Wu, J. Tychsen, M. H. Faber, J. D. Sorensen and **P. Jonathan**. On the distribution of maximum wave and crest height at intermediate water depths. Ocean Eng. 217:107485.

2020: R. Wada, **P. Jonathan**, T. Waseda. Spatial features of extreme waves in the Gulf of Mexico. Proc. 39th Int. Conf. on Ocean, Offshore & Arctic Engineering.

2020: E. Zanini, E. Eastoe, M. J. Jones, D. Randell and **P. Jonathan**. Flexible covariate representations for extremes. Environmetrics 31:e2624.

2019: A. Anokhin, D. Randell, E. Ross and **P. Jonathan**. Spatial and seasonal variability of metocean design criteria in the Southern South China Sea from covariate extreme value analysis, Proc. 38th Int. Conf. on Ocean, Offshore & Arctic Engineering.

2019: G. Feld, D. Randell and **P. Jonathan**. On the estimation and application of directional design criteria, Proc. 38th Int. Conf. on Ocean, Offshore & Arctic Engineering.

2019: G. Feld, D. Randell, E. Ross and **P. Jonathan**. Design conditions for waves and water levels using extreme value analysis with covariates. Ocean Eng. 173:851-866.

2019: R. Shooter, E. Ross, J. A. Tawn and **P. Jonathan**. On spatial conditional extremes for ocean storm severity.

Environmetrics 2019:e2562.

2019: E. Vanem, B. Guo, E. Ross and **P. Jonathan**. Comparing different contour methods with response-based methods for extreme ship response analysis. Mar. Struct. 69:102680.

2019: R. Wada, **P. Jonathan**, T. Waseda and S. Fan. Estimating extreme waves in the Gulf of Mexico using a simple spatial extremes model, Proc. 38th Int. Conf. on Ocean, Offshore & Arctic Engineering.

2018: M. J. Jones, M. Goldstein, **P. Jonathan**, and D. Randell. Bayes linear analysis of sequential optimal design problems. Electron. J. Stat. 12:4002–4031.

2018: M. J. Jones, H. F. Hansen, A. R. Zeeberg, D. Randell, and **P. Jonathan**. Uncertainty quantification in estimation of ocean environmental return values. Coastal Eng. 141:36–51.

2018: E. Ross, S. Sam, G. Feld, D. Randell and **P. Jonathan**. Estimating surge in extreme North Sea storms. Ocean Eng. 154:430–444.

2018: S. Tendijck, E. Ross, D. Randell, and **P. Jonathan**. A model for the directional evolution of severe ocean storms. Environmetrics 30:e2541.

2018: R. Wada, T. Waseda and **P. Jonathan**. A simple spatial model for extreme tropical cyclone seas. Ocean Eng. 169:315-325.

2017: A. Brown, W. Gorter, L. Vanderschuren, P. Tromans, **P. Jonathan**, P. Verlaan. OMAE2017-61005: Design approach for turret moored vessels in highly variable squall conditions. ASME. International Conference on Offshore Mechanics and Arctic Engineering, Volume 3A: Structures, Safety and Reliability, V03AT02A049.

2017: B. Hirst, D. Randell, M. Jones, **P. Jonathan**, B. King and M. Dean. A new technique for monitoring the atmosphere above onshore carbon storage projects that can estimate the locations and mass emission rates of detected sources. Energy Procedia 114: 3716–3728.

2017: P. Northrop, N. Attalides and **P. Jonathan**. Cross-validatory extreme value threshold selection and uncertainty with application to wave heights. J. Roy. Statist. Soc. C, 66:93–120.

2017: E. Ross, M. Kereszturi, M. van Nee, D. Randell and **P. Jonathan**. On the spatial dependence of extreme ocean storm sea states. Ocean Eng. 145:1–14.

2017: E. Ross, D. Randell, K. Ewans, G. Feld and **P. Jonathan**. Efficient estimation of return value distributions from non-stationary marginal extreme value models using Bayesian inference. Ocean Eng. 142:315–328.

2017: R. Towe, E. Eastoe, J. A. Tawn and **P. Jonathan**. Statistical downscaling for future extreme wave heights in the North Sea. Ann. Appl. Stat. 11: 2375-2403.

2016: M. J. Jones, D. Randell, K. Ewans, and **P. Jonathan**. Statistics of extreme ocean environments: non-stationary inference for directionality and other covariate effects. Ocean Eng. 119:30–46.

2016: M. Kereszturi, J. Tawn, and **P. Jonathan**. Assessing extremal dependence of North Sea storm severity. Ocean Eng. 118:242–259.

2016: P. Northrop, **P. Jonathan**, and D. Randell. Threshold modeling of nonstationary extremes. In D. Dey and J. Yan, editors, Extreme Value Modeling and Risk Analysis: Methods and Applications, pages 87–108. Chapman and Hall / CRC.

2016: R. Raghupathi, D. Randell, and **P. Jonathan**. Consistent design criteria for South China Sea with a large-scale extreme value model. Offshore Technology Conference Asia, Kuala Lumpur, OTC-26668.

2016: L. Raghupathi, D. Randell, K. Ewans, and **P. Jonathan**. Fast computation of large scale marginal extremes with multi-dimensional covariates. Comput. Statist. Data Anal. 95:243–258.

2016: L. Raghupathi, D. Randell, K. Ewans, and **P. Jonathan**. OMAE2016-54355: Non-stationary estimation of joint design criteria with a multivariate conditional extremes approach. Proc. 35nd Conf. Offshore Mech. Arct. Eng.

2016: L. Raghupathi, D. Randell, E. Ross, K. Ewans, and **P. Jonathan**. Multi-dimensional predictive analytics for risk estimation of extreme events. (Big Data Foundations Workshop, IEEE High-Performance Computing, Data and Analytics Conference, HIPC2016).

2016: D. Randell, K. Turnbull, K. Ewans, and **P. Jonathan**. Bayesian inference for non-stationary marginal extremes. Environmetrics, 27:439–450.

2016: M. Vogel, J. Hanson, S. Fan, G. Z. Forristall, Y. Li, R. Fratantonio, and **P. Jonathan**. Efficient environmental and structural response analysis by clustering of directional wave spectra. Offshore Technology Conference, Houston, OTC-27039.

2016: R. Wada, T. Waseda, and **P. Jonathan**. Extreme value estimation using the likelihood-weighted method. Ocean Eng. 124:241–251.

2016: Y. Wu, D. Randell, M. Christou, Kevin Ewans, and P Jonathan. On the distribution of wave heights in shallow water. Coastal Eng. 111:39–49.

2015: M. J. Jones, M. Goldstein, **P. Jonathan**, and D. Randell. Bayes linear analysis for Bayesian optimal experimental design. J. Stat. Plan. Inference, 171:115–129.

2015: D. Randell, G. Feld, K. Ewans, and **P. Jonathan**. Distributions of return values for ocean wave characteristics in the South China Sea using directional-seasonal extreme value analysis. Environmetrics, 26:442–450.

2015: R. P. N. Rao, R. Lee, N. Yadav, M. Vahia, **P. Jonathan**, and P. Ziman. On statistical measures and ancient writing systems. Language 91:198–205.

2014: K. Ewans and **P. Jonathan**. Evaluating environmental joint extremes for the offshore industry. J. Marine Syst. 130:124–130.

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Articles in preparation

With Patrik Bohlinger and colleagues: The EnvEx environmental extremes software.

With Marios Christou and Kevin Ewans: On the tri-spectrum for ocean wave characterisation.

With Callum Leach and Kevin Ewans: Practical return value prediction using Bayesian model averages.

With Ed Mackay and Kevin Ewans: A mathematical overview of extreme value modelling in ocean engineering.